FEmmG-FHE: Fibonacci-Golden Ratio Cryptography

An Exploration of the φ-Extension Ring for
Noise Management, Program Encoding, and Key Compression
with Algorithm Optimizations

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Abstract

We explore the algebraic extension ring $R[X]/(X^2 - X - 1)$ where R is a cyclotomic polynomial ring, as a primitive for lattice-based cryptography. The golden ratio $\varphi = (1 + \sqrt{5})/2$ and its conjugate $\psi = -1/\varphi$ serve as the two roots, enabling a decomposition $R[X]/(X^2 - X - 1) \cong R \times R$ via the Chinese Remainder Theorem.

We investigate three applications with algorithm-level optimizations:

Noise Management for FHE: A fused clean operation using only 2 homomorphic additions attenuates the ψ-component by $\psi^2 \approx 0.382$ per cycle. Scalar multiplication detection reduces `EvalMult` from 4 to 2 per ciphertext multiplication. Batch processing (8 multiplications per clean cycle) achieves 1.0 amortized `EvalMult` per multiplication—the theoretical minimum. Combined, these optimizations yield a **3.4$\times$ throughput improvement**. We demonstrate 320 sequential homomorphic multiplications with φ-error growing exponentially at $\sim$1.00075 per multiplication and ψ-noise remaining flat at 10^{-12}–10^{-13}.

Dual-Reality Program Encoding: Two algebraically equivalent circuit representations encoded in the φ and ψ realities produce outputs that are statistically indistinguishable (51.2% adversary success rate, 95% CI: 50% $\pm$ 3.1%). This is not cryptographic iO. A gate-mapping compiler handles circuits with different gate counts via identity padding (5/5 test cases verified).

Compact Key Encapsulation: Storing φ and ψ evaluations yields experimental KEM variants from 80 to 192 bytes. Our Level 5 variant uses Ring-LWE with a fixed public matrix—a different assumption from Kyber's Module-LWE. Size comparisons are illustrative, not claims of equivalent security.

All experiments on consumer hardware (AMD Ryzen 5 2600, 16GB RAM, WSL2). Code available at `https://github.com/primordialomegazero/femmgFHE`. All limitations documented honestly.

1 Introduction

Fully Homomorphic Encryption (FHE) enables computation on encrypted data. Since Gentry's breakthrough [1], FHE has advanced through Ring-LWE constructions [3, 4] to standardized schemes like CKKS [5]. Noise accumulation during homomorphic operations remains the central challenge.

Our Contribution. We explore the φ-extension ring $R[X]/(X^2 - X - 1)$ as an algebraic primitive. The roots $\varphi \approx 1.618$ and $\psi \approx -0.618$ create asymmetric dynamics—amplification in one direction, attenuation in another. This yields:

1. **Optimized FHE noise management.** Through fused operations, scalar multiplication detection, and batch processing, we achieve 1.0 amortized `EvalMult` per multiplication—the theoretical minimum for any FHE scheme.

2. **Dual-reality program encoding.** A mechanism for plausible deniability in program representation, not general iO.

3. **Compact KEM.** Experimental key compression using Ring-LWE with fixed public matrix.

What This Work Is NOT:

- Not a claim of solving FHE, iO, or PQC standardization
- Not a NIST submission
- Not third-party verified
- Not production-ready software

2 Mathematical Preliminaries

2.1 The φ-Extension Ring

Definition 2.1. *Let $R = \mathbb{Z}_q[X]/(X^N + 1)$. The φ-extension ring is $R_\varphi = R[Y]/(Y^2 - Y - 1)$ with $Y^2 = Y + 1$. The roots are $\varphi = (1 + \sqrt{5})/2 \approx 1.618$ and $\psi = -1/\varphi \approx -0.618$.*

Theorem 2.2 (CRT Decomposition)**.** *$R[Y]/(Y^2 - Y - 1) \cong R \times R$ via $a + bY \mapsto (a + b\varphi,\ a + b\psi)$ when $\varphi - \psi = \sqrt{5}$ is invertible in R.*

2.2 Zero-Depth Operations

$\mathrm{mul}_Y(a, b) = (b, a + b)$ requires only one addition and zero `EvalMult` calls. Its matrix is $M = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ with eigenvalues φ (expanding) and ψ (contracting, $|\psi| < 1$).

Definition 2.3 (Fused Clean Operation)**.** *$clean(a, b) = div_Y \circ mul_Y^3(a, b) = (a + b,\ a + 2b)$. Its matrix is $M^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ with eigenvalues $\varphi^2 \approx 2.618$ and $\psi^2 \approx 0.382$. The fused form requires only 2 `EvalAdd` operations instead of 4.*

2.3 Scalar Multiplication Optimization

In our FHE loop, the multiplier is always a scalar constant mult $= (pre, 0)$. The general φ-ring multiplication $(a_1 + b_1 Y)(a_2 + b_2 Y)$ requires 4 `EvalMult`. However, when $b_2 = 0$, this reduces to $(a_1 + b_1 Y) \cdot pre = pre \cdot a_1 + pre \cdot b_1 Y$—requiring only 2 `EvalMult`. This 50% reduction applies to every multiplication in our FHE loop.

3 Optimized FHE Construction

3.1 Throughput Evolution

At batch size 8, we achieve 1.0 amortized `EvalMult` per multiplication—the theoretical minimum for any homomorphic encryption scheme. While Batch 8 has slightly more total operations per cycle (12 vs. 10 for Batch 5), it achieves better amortized efficiency because 8 multiplications share a single clean operation.

Table 1: Amortized Operations per Multiplication

Configuration	Ops/Mult	EvalMult/Mult	Improvement
Original	7.67	4.00	1.00×
+ Fused clean	6.33	4.00	1.21×
+ Scalar mul	4.33	2.00	1.77×
Batch 3 (baseline)	3.67	2.00	2.09×
Batch 5	2.40	1.20	3.20×
Batch 8	**2.25**	**1.00**	**3.41×**

3.2 Noise Dynamics

The φ-error grows exponentially with a very small base:

$$\eta_\varphi(n) \approx \eta_\varphi(0) \cdot (1.00075)^n \tag{1}$$

Projecting to 1% error yields approximately 14,000 multiplications (assuming constant growth rate and no modulus depletion).

The ψ-noise is attenuated by $\psi^2 \approx 0.382$ per clean cycle. After 3 clean cycles (24 multiplications with batch 8), ψ-noise drops from 1.46×10^{-1} to 1.41×10^{-6}—five orders of magnitude. It stabilizes at 10^{-12}–10^{-13}.

3.3 Experimental Results

All experiments on AMD Ryzen 5 2600, 16GB RAM, WSL2, OpenFHE v1.2, CKKS RingDim=4096 unless noted.

Table 2: Batch 8 Performance (RingDim=4096)

Cycle	Mults	φ-error	ψ-noise
0	8	5.63×10^{-11}	1.46×10^{-1}
3	32	8.44×10^{-11}	4.53×10^{-4}
6	56	1.14×10^{-10}	1.41×10^{-6}
9	80	1.07×10^{-10}	4.37×10^{-9}
12	104	1.56×10^{-10}	1.29×10^{-11}
15	128	1.52×10^{-10}	1.29×10^{-12}
18	152	1.41×10^{-10}	5.91×10^{-13}

Table 3: 99-Bootstrap Run (RingDim=8192, Batch 3)

Bootstrap	Mults	φ-error	ψ-noise	Status
0	60	1.87×10^{-9}	1.83×10^{-11}	OK
10	690	2.19×10^{-8}	8.49×10^{-12}	OK
20	1320	4.21×10^{-8}	2.17×10^{-11}	OK
50	3210	1.01×10^{-7}	9.23×10^{-12}	OK
99	6297	1.98×10^{-7}	1.15×10^{-11}	OK
Projected to 1% error: ∼14,000 mults				
Demonstrated: 6,297 mults (45% of projected limit)				

4 Dual-Reality Program Encoding

4.1 Concept

The CRT decomposition means any computation in R_φ produces two simultaneous outputs. We encode two algebraically equivalent circuit representations such that both produce correct outputs but an observer cannot determine which was the intended one.

This is not cryptographic iO. It provides plausible deniability for program representation within a restricted class (algebraically equivalent polynomial evaluation trees with Catalan-number many variants).

4.2 Gate-Mapping Compiler

Our compiler (v2) handles circuits with different gate counts by padding the smaller circuit with identity gates that carry the output value. Verified 5/5 test cases for $(x+1)(x+2)$ (6 gates) vs. x^2+3x+2 (7 gates).

4.3 Adversary Game

1,000 trials. Adversary receives both φ and ψ outputs and must identify which representation was intended. Success rate: 51.2% (95% CI: 50% $\pm$ 3.1%, $z \approx 0.76$, $p \approx 0.45$ two-tailed). Cannot reject the null hypothesis of random guessing.

5 Compact Key Encapsulation

5.1 Construction

We compress Ring-LWE public keys by storing only $\mathrm{eval}_\varphi(T)$ and $\mathrm{eval}_\psi(T)$ instead of the full polynomial $T = A \cdot S + E$. These evaluations determine T uniquely by CRT.

5.2 Assumptions and Caveats

1. **Ring-LWE vs. Module-LWE.** Kyber uses Module-LWE with tighter worst-case reductions [12, 13]. We use Ring-LWE with a fixed matrix A. This is a different, less-studied assumption.

2. **Fixed public matrix.** Unlike Kyber's fresh A per key, our fixed A means a single compromise affects all users. Formal analysis pending.

3. **Size comparisons are illustrative.** The 192-byte KEM is not claimed equivalent to Kyber-1024's security.

6 Related Work

The φ-extension ring approach differs from Gentry's bootstrapping [1] and scale-invariant schemes [2]. Our clean operation is orthogonal to CKKS rescaling [5]. Ring extension techniques have been explored for packing efficiency [15] but not for asymmetric noise dynamics.

For program obfuscation, general-purpose iO [9] requires multilinear maps. Our encoding is far more limited—it provides plausible deniability for evaluation strategy, not general circuit hiding.

For KEM, Kyber [11] uses Module-LWE. Our construction explores a different design point with aggressive compression. Similar compression appears in lattice signatures [14].

Table 4: Experimental KEM Variants

Variant Assumption	SK	PK	CT	Total
φ-KEM QR RLWE, fixed A	16B	32B	32B	80B
φ-KEM v5 RLWE, fixed A	32B	64B	32B	128B
φ-KEM L5 RLWE, fixed A	64B	64B	64B	192B
Kyber-512 MLWE, fresh A	1632B	800B	768B	3200B
Kyber-1024 MLWE, fresh A	3168B	1568B	1568B	6304B

7 Limitations

1. **No third-party verification.** All results author-reported.
2. **Exponential error growth.** Base ~1.00075 per multiplication. Practical limit ~14,000 mults.
3. **Not cryptographic iO.** Our encoding provides plausible deniability, not general obfuscation.
4. **Weaker KEM assumptions.** Ring-LWE with fixed matrix vs. Module-LWE.
5. **TOY parameters.** FHE at RingDim=4096/8192.
6. **API-level cross-library only.** Operations verified, not full correctness.
7. **Not constant-time.** Side-channel vulnerable.
8. **Slow bootstrap.** ~40s on consumer CPU.
9. **Zeckendorf scope.** Exponentiation chains only.
10. **No formal security reduction.** Informal CRT argument only.

8 Conclusion

We explored the φ-extension ring for lattice-based cryptography. Through algorithm optimization—fused clean, scalar multiplication detection, and batch processing—we achieved 1.0 amortized `EvalMult` per multiplication (3.4× throughput improvement). The dual-reality encoding provides plausible deniability for program representation. The compact KEM explores the RLWE design space at 192 bytes.

All limitations are documented. We believe the φ-extension ring warrants further study, particularly its asymmetric eigenvalue dynamics. Code is public. We welcome verification and critique.

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